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Executive incentives under common ownership

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ABSTRACT

Relative performance evaluation (RPE) increases competition and limits pay-for-luck by rewarding executives for outperforming rivals. This study tests whether institutional investors reduce RPE use when they own stakes in competing firms. Contrary to this, the Big Three asset managers – BlackRock, Vanguard, and State Street – demonstrate strong preferences for RPE, reflected in portfolio firms' RPE adoptions, say-on-pay vote support, and peer group selections. No evidence suggests that common ownership by these or other institutional investors reduces RPE, as confidence bounds and point estimates are near zero. Overall, the rising prevalence of RPE challenges concerns about anticompetitive effects from common ownership.

1. Introduction

Large, diversified institutional shareholders have become increasingly prominent in the United States' financial markets. The 'Big Three' asset management firms – BlackRock, Vanguard, and State Street – own substantial stakes in most publicly traded companies and consequently hold positions in firms that are natural product-market rivals.[1] Such 'common owners' may wish to discourage competition and even foster collusion among their portfolio firms to extract rents above those implied by a competitive market equilibrium (Bresnahan and Salop, 1986; Reynolds and Snapp, 1986). This paper tests whether institutional investors influence executive compensation in a manner consistent with achieving anti-competitive outcomes, specifically reducing relative performance evaluation (RPE), and finds clear evidence to the contrary.

Seminal work on contract theory contends that an agent's performance should not be measured in a vacuum, but relative to the performance of peers that operate under similar circumstances (Lazear and Rosen, 1981; Holmstrom, 1982; Green and Stokey, 1983). This literature shows that principals and risk-averse agents can achieve Pareto improvement by incorporating RPE into compensation arrangements.[2] However, the foundational theories supporting RPE assume that each principal's utility is based solely on the

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[1] Fichtner et al. (2017) report that one of the Big Three is the largest shareholder in 88 % of S&P 500 companies.

[2] RPE identifies performance due to an agent's effort by filtering out aggregate market or industry trends. RPE schema regularly used in executive compensation include evaluating performance net of the return to a benchmark group or index and rank-order tournaments where rewards are based on percentile achievement of stock returns or various accounting metrics *versus* a peer group.

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performance of one agent. In contrast, it may not be advantageous for principals with ownership interests in multiple, competing agents to adopt RPE-based incentive contracts (Kraus and Rubin, 2010).

Indeed, theory shows that RPE can intensify competition and even incite destructive sabotage (Lazear, 1989; Gibbons and Murphy, 1990). Recent empirical studies support this idea, consistently concluding that RPE heightens competition among rival firms.[3] Since common owners have stakes in competing firms and could profit from collusion, they may seek to discourage or abolish RPE among their portfolio constituents.

This paper examines whether common owners reduce RPE and finds no evidence to support this across three distinct dimensions. First, firms with higher levels of common ownership are no less likely to use RPE in executive compensation. Second, common owners do not exhibit differential support for RPE in 'say-on-pay' shareholder referendums. Third, firms do not avoid benchmarking RPE against peers that share a common owner with them. In all settings, the point estimates for the effects of common ownership are statistically and economically small or oppositely signed, contradicting the hypothesis that common owners discourage RPE to make collusion incentive-compatible for their firms' managers.

To test this theory, I begin with two independent variables at the firm-year level: *Institutional Direct Ownership* is the fraction of outstanding shares held by institutional investors, and *Institutional Common Ownership* is the fraction of other firms in the same industry that share an institutional blockholder in common with the firm.[4] The first measure captures direct ownership by institutions with over $100 million in assets as reported to the SEC in Form 13F, and the second captures the extent to which these institutions also own the firm's competitors. Using detailed data on executive compensation awards from Incentive Lab, I build a dummy variable, *RPE*, that flags if the firm-year uses RPE and a continuous variable, *RPE%*, that equals the dollar-weighted fraction of CEO pay linked to RPE awards.

I begin by regressing the RPE outcomes on the ownership measures. I find no effect of *Institutional Direct Ownership* on RPE, but surprisingly, I find a strong, positive effect of *Institutional Common Ownership* on RPE. These results are inconsistent with collusion, but because RPE increases competitive pressures, it is unclear why common ownership would increase RPE.

To understand why *RPE* loads positively on aggregate common ownership, I decompose the two ownership measures into those associated with the Big Three asset managers and those attributable to other institutional investors. The Big Three are central to the ongoing debate on common ownership because in addition to their diversified holdings, their voting power grants them considerable sway over corporate policies, including executive compensation.[5] Additionally, the Big Three use dedicated stewardship teams to engage directly with portfolio companies and influence governance practices behind the scenes.[6] The Big Three may demand RPE for its pay-for-performance properties, as their size uniquely positions them to hold managers accountable. Alternatively, they may advocate for RPE because, as large asset managers subject to significant scrutiny, they seek to avoid any appearance of collusion or conflicts of interest.

I define *Big 3 Direct Ownership* as the fraction of shares held by the Big Three asset managers, and I define *Big 3 Common Ownership* as the fraction of other firms in the same industry that share a Big Three blockholder in common with the firm. I build analogous measures for non-Big Three institutional investors: *Other Direct Ownership* and *Other Common Ownership*. Regressing *RPE* on these four refined measures produces a strong, positive coefficient on *Big 3 Direct Ownership* and insignificant coefficients on the remaining measures. The positive effects of the former are the opposite of what a collusion hypothesis would predict, and the statistically insignificant coefficients on the latter are inconsistent with the idea that common owners discourage competition. The fact that *Big 3 Direct Ownership* loads positively while *Big 3 Common Ownership* is statistically indistinguishable from zero suggests that these asset managers favor RPE for its ability to align pay with performance, rather than to avoid apparent conflicts of interest among their commonly owned companies.

To establish causality and address the possibility that ownership and compensation are endogenously determined, I use an instrumental variables specification that exploits the fact that much of the Big Three's holdings are mechanically determined by passive ownership that tracks major stock index lists. I instrument *Big 3 Direct Ownership* with a dummy variable that flags whether the firm is an S&P 500 index constituent, and I instrument *Big 3 Common Ownership* with a variable equal to the fraction of other firms in the same industry that are S&P 500 members. Since S&P 500 membership is primarily driven by firm size, I argue that these instruments plausibly satisfy the exclusion restriction (see section 3.1 below).

In first-stage regressions, I find strong relationships between the instruments and the respective endogenous variables even in the presence of firm fixed effects, which satisfies the relevance condition. In 2SLS, the instrumented effects are consistent with the OLS

[3] Feichter et al. (2022) find that RPE emboldens managers into taking more frequent and complex competitive actions against peers, such as introducing new products and launching new marketing campaigns. Bloomfield et al. (2024) find that firms strategically disclose information that harms their RPE peers' stock prices. Schneider et al. (2024) find that banks are less likely to invest in loans underwritten by rival banks named as RPE peers.

[4] I define a blockholder using the traditional 5 % ownership cutoff (*e.g.*, Holderness, 2009). For robustness, I alternatively measure common ownership using O'Brien and Salop's (2000) modification to the Herfindahl-Hirschman industry concentration index that captures partial common, ΔMHHI, as well as a measure of profit-weighted common ownership following Harford et al. (2011) and find consistent results. Industry is by four-digit *SIC* code, though the results are robust to using the Hoberg-Phillips industry classifications. Please see section 4.5 below and the Internet Appendix for tabulation of robustness tests.

[5] Bebchuk and Hirst (2019) estimate that the Big Three collectively cast 25 % of all votes at S&P 500 companies, and they extrapolate that this may rise to over 40 % of all votes at those companies within two decades.

[6] For instance, the *Wall Street Journal* reported that in 2021, Blackrock's stewardship team "engaged with 2300 companies *via* emails, phone calls and meetings," "weighing in on matters as varied as executive compensation, climate change and abortion access" (Au-Yeung, 2022).

results: *Big 3 Direct Ownership* positively relates to RPE while the effects of *Big 3 Common Ownership* are small and statistically indistinguishable from zero with tight confidence bounds. These results firmly refute the idea that the Big Three disfavor RPE in executive compensation to encourage collusion.

Next, I study how common ownership and RPE affect shareholder advisory votes on executive compensation known as 'say-on-pay'. If common owners wish to facilitate collusion, they may use say-on-pay to voice support for compensation that does not include RPE. However, Appendix 2 reports an excerpt from Vanguard's recent U.S. Proxy Voting Policy, in which Vanguard lists the alignment of pay and relative performance as the first consideration when evaluating executive compensation. Specifically, Vanguard states that it mainly assesses this through "analysis of three-year total shareholder return and realized pay over the same period vs. a relevant set of peer companies." Including relative metrics in pay awards directly remedies this by increasing (decreasing) realized pay following outperformance (underperformance) of peers. Vanguard further states that a warning sign is raised if "The plan uses absolute metrics only." Thus, these voting guidelines communicate a clear preference for RPE in executive pay, and funds do not typically deviate from their voting guidelines (Couvert, 2021).

I obtain firm-level say-on-pay vote tallies from ISS Voting Analytics and define *Votes* as the percentage of votes cast in favor of say-on-pay. Although BlackRock, Vanguard, and State Street are inherently distinct institutions, their voting is significantly correlated due to factors including their overwhelming coincidence of interests (Bebchuk and Hirst, 2022), their shared philosophy concerning corporate governance (Bubb and Catan, 2022), and their overlapping investor ideology (Bolton et al., 2020). As such, I continue to focus on measures that separate the Big Three's ownership (*i.e.*, *Big 3 Direct Ownership* and *Big 3 Common Ownership*) from ownership by other institutional investors (*i.e.*, *Other Direct Ownership* and *Other Common Ownership*). This approach is consistent with Bebchuk and Hirst (2022), who state, "As a result, it makes sense to aggregate the voting power of the Big Three in order to properly understand their power."

I regress *Votes* on the four ownership measures and their interactions with the RPE variables. The hypothesis that the common owners support pay without RPE predicts negative coefficients on these interaction terms, but that is not what I find. Instead, the coefficient on *Big 3 Direct Ownership* × *RPE* is positive while the coefficients on the remaining interaction terms are statistically and economically insignificant. Taken together, the Big Three reveal a unique preference for RPE through their voting behavior, which is not mitigated by their common ownership, and other institutions appear indifferent to RPE in their votes. These results hold in alternative specifications that additionally interact the ownership measures with other dimensions of pay, including the log of total compensation and the proportion of pay delivered in stock and option grants.

Last, I test whether common owners attempt to circumvent the competitive effects of RPE by altering the composition of underlying award reference groups. Redesigning and recalibrating an entire compensation program to exclude RPE could be prohibitively challenging and costly. Instead, the common owners could simply pressure firms to avoid peer groups that pit their portfolio firms in direct contest with each other. This could manifest as benchmarking RPE against a general market index that is too broad to capture the gains from industry-level collusion or against a custom peer group that excludes commonly held rivals by design. In tests of the former, I find no evidence that common ownership alters benchmark group choice.

To test the latter, I build an annual firm-peer pair variable, *RPE Peer*, that flags if the benchmarking firm chooses the other firm as an RPE peer that year. I define pairwise variables, *Big 3 Common Block* and *Other Common Block,* as dummies that indicate if the two firms share one of the Big Three or one of the non-Big Three institutions in common as a blockholder, respectively. Regressing *RPE Peer* on these measures of common ownership produces small but positive effects for the former and insignificant effects for the latter, both inconsistent with avoiding competition. I then interact the common ownership measures as well as the direct ownership measures (at the benchmarking firm) with a measure of how closely the firms compete. To measure competition, I use the pairwise product similarity score developed by Hoberg and Phillips (2010, 2016), which measures the extent of product market overlap. Firm-pairs that score higher on this metric compete to offer similar goods and thus stand the most to gain from collusion. The coefficients of *RPE Peer* on *Big 3 Direct Ownership* × *Competition* are large and positive even in the presence of pairwise fixed effects. This implies that after a firm experiences an increase in Big Three ownership, the firm tends to benchmark RPE against closer competitors. Interacting *Competition* with *Big 3 Common Block* or the non-Big Three measures produces coefficients that are statistically near zero. Thus, common owners do not avoid RPE that rewards outperforming their other portfolio companies. Additionally, I define a market-based measure of competitive overlap using pairwise stock return correlations and find similar effects.

This paper contributes to the debate on the anticompetitive effects of common ownership by showing that common owners do not eschew RPE, which intensifies competition. The statistically insignificant results on common ownership and RPE align with those on common ownership and product market competition in Koch et al. (2021) and Lewellen and Lowry (2021). Although those papers do not study compensation, they carefully address various methodological issues and find no consistent anticompetitive effects of common ownership in the broad economy, in contrast to empirical literature that reaches mixed conclusions in narrower settings.[7] My paper complements theirs by eliminating RPE as a potential channel through which common owners might attempt to achieve anticompetitive outcomes.

In the same vein, my paper adds to the literature on how common ownership shapes executive incentives. First, the paper

[7] Within the airline industry, Azar et al. (2018) find that common ownership has anticompetitive effects on ticket prices, while Kennedy et al. (2017) and Dennis et al. (2022) find no such effects. Within banking, Azar et al. (2022) find that common ownership limits competition regarding interest rates and fees, while Gramlich and Grundl (2017) show that these effects are economically small and of mixed sign. Regarding mergers and acquisitions, Matvos and Ostrovsky (2008) find that common ownership may explain why shareholders agree to overpay, while Harford et al. (2011) show that this result does not hold when common ownership is measured by relative weights.

demonstrates the importance of disentangling the effects of common ownership from the direct effects of ownership by large stewardship-focused asset managers that happen to be common owners. By introducing this separation, my paper helps clarify results from Kwon (2016), who finds that common ownership is positively related to RPE, and Bloomfield et al. (2024), who find that common ownership and RPE are positively related or unrelated depending on specification.[8] Specifically, my paper shows that the seemingly positive effect of *All Common Ownership* on RPE is in fact driven entirely by the Big Three's direct ownership. Common ownership by the Big Three and other institutions has no discernable effect on RPE. My result that the Big Three's direct ownership increases RPE corroborates the consensus of the extant compensation and governance literatures that institutional blockholders are competent monitors of executive activity and serve an important role in motivating and disciplining managers (Edmans and Holderness, 2016; Edmans et al., 2017).

Additionally, by testing for *explicit* RPE as disclosed in mandatory SEC regulatory filings, my paper contrasts with those that study *implicit* RPE, which only infers RPE through regression analysis.[9] Papers that study common ownership and implicit RPE include Anton et al. (2016) and Liang (2016), who find that common ownership decreases implicit RPE, and Kwon (2016) who finds the opposite.[10] Testing for implicit RPE requires assumptions including peer group composition, performance metrics, and payout style that may induce measurement error and cloud inference (Murphy, 1999; Bannister and Newman, 2003). Importantly, implicit tests will fail to detect RPE even if an executive's entire realized pay is contingent upon it unless the target award value also covaries with peer performance.[11] Using explicit RPE data from Incentive Lab allows me to observe contract details and conclude that common ownership is not negatively related to RPE.

Furthermore, studying explicit RPE allows me to observe variation in executive incentive design that is not captured by measures of executives' wealth-performance sensitivity (WPS), which quantifies how executive wealth changes with firm value, as calculated by functions of stock- and option-based pay.[12] Although Anton et al. (2023) find that common ownership lowers WPS in a manner that is potentially consistent with an anticompetitive effect, WPS inherently ignores whether the underlying pay is delivered as RPE-based awards. Cash-, stock-, and option-based awards may all be contingent upon or scale with relative performance, and thus WPS is unable to capture RPE or its competitive effects.[13] My results show that the largest common owners – the Big Three – prefer RPE at their portfolio firms, intensifying competitive pressures, while other common owners do not have anticompetitive effects.

Last, this paper contributes to an emerging literature on how the Big Three shape corporate outcomes. The Big Three can potentially have large effects on firms' policies because their large ownership stakes in portfolio firms grant them high propensities to be pivotal voters in shareholder votes (Bebchuk and Hirst, 2019; Griffin, 2020). Additionally, Brav et al. (2024) note that the Big Three have greater access to management through their prominent shareholder status and can shape firms' outcomes through direct communication. Azar et al. (2021) show that Big Three ownership reduces carbon emissions, consistent with the idea that their common ownership may help resolve certain collective action issues. Gormley et al. (2023) find that the Big Three's activism efforts increased the number of female directors on boards following the #MeToo movement. To understand why the Big Three and other passive investors engage as shareholders, Lewellen and Lewellen (2022) demonstrate that these institutions derive substantial benefits from stock price appreciation, primarily through assets under management fees and fund flows generated by their activism. Brav et al. (2023) survey the literature on passive ownership and show that the benefits from engagement are higher for the Big Three than other institutions due to economies of scale. My paper adds to this literature by showing that Big Three ownership increases RPE use – while non-Big Three ownership does not – and importantly, common ownership does not offset this effect.

2. Institutional example

The United States' airline industry has been the focus of academic debate regarding the anticompetitive effects of common ownership. The industry is dominated by a small number of large companies that compete to sell mostly undifferentiated products, and the companies in this industry are highly connected by their institutional shareholders. Researchers have speculated that common ownership has caused airlines to collude, resulting in artificially high ticket prices. However, the empirical evidence of such effects is

[8] Although Bloomfield et al. (2024) briefly examine RPE, their main contribution is that they find no relationship between common ownership and revenue-based pay. My paper complements theirs by showing that, contrary to the collusion hypothesis, the Big Three prefer RPE in executive compensation.

[9] The canonical *implicit* RPE test regresses executive pay on firm and industry returns; a positive coefficient on firm return and a negative coefficient on industry return is considered evidence of implicit RPE.

[10] These papers augment the usual implicit RPE test by interacting firm and industry returns with measures of industry concentration such as the Herfindahl-Hirschman index (HHI), following the approach in Aggarwal and Samwick (1999), and the increase in industry concentration implied by partial common ownership captured by O'Brien and Salop's (2000) modified HHI (ΔMHHI). In this augmented test, a positive coefficient on the interaction of industry return and ΔMHHI suggests that firms with greater partial common ownership use less implicit RPE.

[11] For example, consider an executive who, every year, receives her entire compensation package as an explicit RPE contract with a target value of $1 million. Since the U.S. Securities and Exchange Commission (SEC) requires only the fair value of the award to be incorporated into total flow compensation, the implicit test will fail to detect any RPE because the fixed value of $1 million does not vary with performance even though the award payout is entirely determined by relative performance. See SEC Final Rule 33-8732a, November 7, 2006.

[12] See Hall and Liebman (1998), Jensen and Murphy (1990), and Edmans et al. (2009), for the development of WPS.

[13] Furthermore, since RPE awards often deny grants for underperformance (*e.g.*, no stock payout for performance below the 25th percentile), the Big Three's preference for RPE may partially explain Anton et al.'s (2023) result that common ownership reduces WPS if RPE has reduced stock compensation at underperforming firms.

Table 1
Institutional example: U.S. Airlines.

Panel A: RPE Use

	Passengers	RPE	RPE%
American Airlines Group, Inc.	199,640,000	Yes	27.8 %
Delta Air Lines, Inc.	186,390,000	Yes	18.9 %
Southwest Airlines Co.	157,677,000	No	
United Continental Holdings, Inc.	148,067,000	Yes	27.1 %
Alaska Air Group, Inc.	44,034,000	Yes	18.3 %
JetBlue Airways, Inc.	40,038,000	Yes	37.4 %

Panel B: Shares Held by Largest 5 Investors

American Airlines Group, Inc.		**Delta Air Lines, Inc.**		**Southwest Airlines Co.**	
T. Rowe Price Assoc., Inc.	13.90 %	Blackrock Inc.	9.10 %	PRIMECAP Mgmt Co.	11.80 %
Berkshire Hathaway Inc.	9.00 %	Berkshire Hathaway Inc.	8.00 %	Blackrock Inc.	7.60 %
PRIMECAP Mgmt Co.	8.90 %	Vanguard Group, Inc.	6.10 %	Berkshire Hathaway Inc.	7.00 %
Blackrock Inc.	6.80 %	JPMorgan Chase & Co.	4.50 %	Vanguard Group, Inc.	6.20 %
Vanguard Group, Inc.	6.00 %	State Street Co.	4.20 %	Fidelity Mgmt & Co.	5.60 %
United Continental Holdings, Inc.		**Alaska Air Group, Inc.**		**JetBlue Airways, Inc.**	
Blackrock Inc.	9.50 %	T. Rowe Price Assoc., Inc.	10.10 %	Vanguard Group, Inc.	8.00 %
Berkshire Hathaway Inc.	9.10 %	Vanguard Group, Inc.	9.70 %	Fidelity Mgmt & Co.	7.90 %
Vanguard Group, Inc.	6.80 %	Blackrock Inc.	6.20 %	Blackrock Inc.	7.80 %
PRIMECAP Mgmt Co.	6.20 %	PRIMECAP Mgmt Co.	5.20 %	PRIMECAP Mgmt Co.	5.90 %
PAR Capital Mgmt, Inc.	5.10 %	PAR Capital Mgmt, Inc.	3.70 %	Goldman Sachs & Co.	3.80 %

Panel C: RPE Peer Group Constituents

American Airlines Group, Inc.	**Delta Air Lines, Inc.**	**Southwest Airlines Co.**
Alaska Air Group, Inc.	Alaska Air Group, Inc.	*N/A*
Delta Air Lines, Inc.	American Airlines Group, Inc.	
JetBlue Airways, Inc.	JetBlue Airways, Inc.	
Southwest Airlines Co.	Southwest Airlines Co.	
Spirit Airlines, Inc.	United Continental Holdings, Inc.	
United Continental Holdings, Inc.		
United Continental Holdings, Inc.	**Alaska Air Group, Inc.**	**JetBlue Airways, Inc.**
Alaska Air Group, Inc.	Air Canada	Alaska Air Group, Inc.
American Airlines Group, Inc.	Allegiant Travel Co.	American Airlines Group, Inc.
Delta Air Lines, Inc.	American Airlines Group, Inc.	Delta Air Lines, Inc.
JetBlue Airways, Inc.	Delta Air Lines, Inc.	Hawaiian Holdings, Inc.
Southwest Airlines Co.	Hawaiian Holdings, Inc.	Southwest Airlines Co.
	JetBlue Airways, Inc.	Spirit Airlines
	Republic Airways Holdings, Inc.	United Continental Holdings, Inc.
	SkyWest Airlines, Inc.	
	Southwest Airlines Co.	
	Spirit Airlines	
	United Continental Holdings, Inc.	
	WestJet Airlines, Ltd.	

Data as of FY2017.

mixed: Azar et al. (2018) report that common ownership has anticompetitive effects on airline ticket prices, while Kennedy et al. (2017) and Dennis et al. (2022) find no such effects. In this section, I informally explore the likelihood that the Big Three and other common owners have softened competition by reducing RPE in airline executive compensation.

Four major companies command the majority of market share in the United States airline industry. Panel A of Table 1 lists the largest U.S. airlines ranked by number of annual passengers in sample year 2017. American Airlines Group (American), Delta Air Lines (Delta), Southwest Airlines (Southwest), and United Continental Holdings (United) each accounted for over 140 million passengers that year. The next biggest airlines are much smaller: Alaska Air Group (Alaska) and JetBlue Airways (JetBlue) had over 40 million passengers.

Panel B of Table 1 lists each major airline's top 5 largest shareholders, revealing that the airlines share a substantial amount of common ownership. The two biggest of the Big Three, BlackRock and Vanguard, are top shareholders in all six major airlines. Absent agency issues, each airline will seek to maximize its individual value, including through competitive actions that benefit it at the expense of other airlines. However, since common owners are exposed to the gains and losses of all their portfolio constituents, they

Table 2
Summary statistics.

	Mean	Std.Dev.	25th	Median	75th
Relative Performance Evaluation					
RPE	0.472	0.499	0.000	0.000	1.000
RPE%	0.101	0.146	0.000	0.000	0.166
Ownership and Instruments					
All Direct Ownership	0.725	0.233	0.627	0.781	0.883
All Common Ownership	0.544	0.351	0.246	0.623	0.867
Big 3 Direct Ownership	0.160	0.073	0.121	0.166	0.208
Big 3 Common Ownership	0.518	0.363	0.145	0.589	0.857
Other Direct Ownership	0.566	0.196	0.466	0.595	0.697
Other Common Ownership	0.111	0.192	0.000	0.000	0.143
Firm in S&P 500	0.418	0.493	0.000	0.000	1.000
Industry in S&P 500	0.540	0.291	0.396	0.600	0.764
Industry ΔMHHI	0.535	0.306	0.357	0.498	0.616
Industry HHI	0.213	0.169	0.085	0.170	0.286
Firm Characteristics					
Industry Exposure	0.411	0.235	0.218	0.411	0.601
Stock Return	0.184	0.405	−0.051	0.136	0.347
Return-on-Assets	0.084	0.103	0.033	0.074	0.131
Book-to-Market	0.462	0.412	0.205	0.390	0.653
Leverage	0.273	0.211	0.099	0.250	0.399
log MCAP	8.760	1.407	7.881	8.667	9.643
Multi-segment	0.858	0.349	1.000	1.000	1.000
Executive Characteristics					
CEO/Chair Duality	0.384	0.486	0.000	0.000	1.000
General Ability Index	−0.000	1.000	−0.780	−0.245	0.556
Executive Age	56.385	6.706	52.000	56.000	60.000
Executive Tenure	7.365	5.287	3.000	6.000	11.000
Say-on-Pay					
Votes	0.902	0.124	0.894	0.949	0.972
log Executive Pay	15.757	0.882	15.329	15.856	16.323
Pct Stock-based Pay	0.552	0.238	0.435	0.592	0.716
Benchmark Group, Conditional on using RPE					
Relative to Peer Group	0.552	0.497	0.000	1.000	1.000
Relative to Industry Index	0.184	0.387	0.000	0.000	0.000
Relative to Market Index	0.341	0.474	0.000	0.000	1.000

This table presents summary statistics at the firm-year level. The sample consists of all firms in the Incentive Lab database from 2010 to 2021 with non-missing data items after merging with BoardEx, Compustat, CRSP, and Thomson/Refinitiv. The sample contains 11,516 observations while the sub-sample conditional on using RPE is 5441 observations. Variables are defined in Appendix 1. Continuous variables are winsorized at the 1st and 99th levels.

may wish for airlines to internalize certain externalities and collude to maximize their total portfolio value instead.

To achieve incentive-compatibility, common owners will avoid compensation that rewards airline executives for outperforming other airlines. However, as Panel A shows, five of the six major airlines use RPE in their executive compensation programs. RPE represents an economically large fraction of pay, accounting for 27 % of pay at American and United, 18 % at Alaska and Delta, and 37 % at JetBlue. Furthermore, these airlines all choose to benchmark RPE awards against each other.[14] For example in Panel C, American names Alaska, Delta, JetBlue, Southwest, Spirit, and United as its comparator group. That is, American's CEO gets paid *more* when his airline outperforms the others despite connections through common ownership.

The data show that airline executives have substantial monetary incentives to compete against each other. Given the prevalence of RPE and the high common ownership at the airlines, it is unlikely that common owners have sought to encourage collusion by discouraging relative incentives.

[14] Alternatively, common owners could broaden the RPE peer group by adding companies from outside the airline industry, avoid competition by excluding commonly held airline peers, or use a broadly defined index (*e.g.* the S&P 500) that does not perfectly reflect benefits from airline industry collusion.

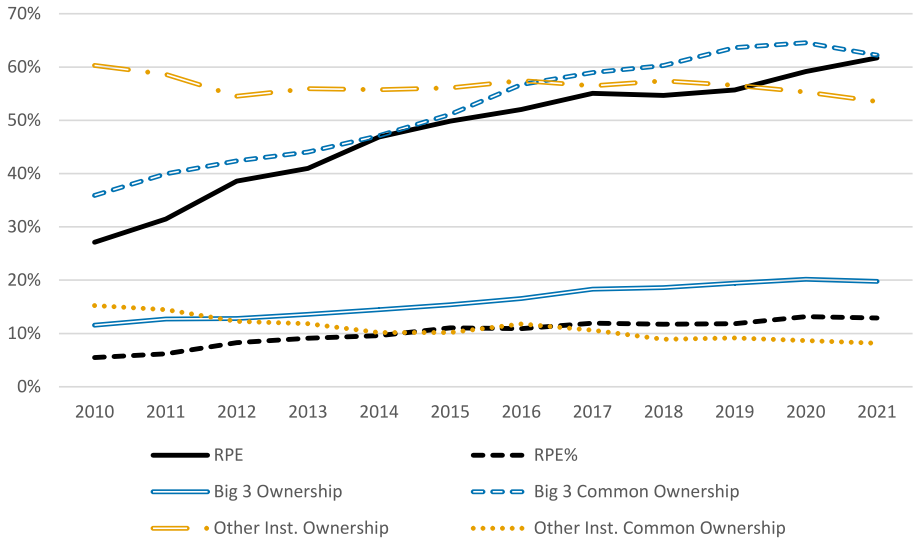

Fig. 1. RPE and common ownership.
This table plots the prevalence of RPE use in executive compensation alongside institutional ownership and common ownership. *RPE* is a dummy variable equal to one if the firm uses RPE in CEO compensation. *RPE%* is the dollar fraction of CEO compensation tied to RPE awards. *Big 3 Direct Ownership* is the fraction of shares owned by BlackRock, State Street, and Vanguard. *Big 3 Common Ownership* is the fraction of same-industry firms that share a common Big Three blockholder with the firm. *Other Direct Ownership* is the fraction of the firm's stock owned by institutional investors excluding the Big Three. *Other Common Ownership* is the fraction of same-industry firms that share at least one non-Big Three institutional investor as a blockholder with the firm. The sample consists of all firms in the Incentive Lab database with non-missing data items after merging with BoardEx, Compustat, CRSP, and Thomson/Refinitiv.

3. Data and empirical design

To test the idea that the common owners discourage RPE among portfolio firms, I develop hypotheses and build models in three distinct settings: (1) firm-level RPE use, (2) voting in say-on-pay referendums, and (3) RPE benchmark choice and peer selection.

3.1. RPE use

The paper's core thesis is that if common owners wish to soften competition or encourage coordination among portfolio firms that are otherwise natural product market competitors, they will avoid compensation plans that explicitly reward executives for out-performing competitors. Formally:

Hypothesis 1. *If common owners discourage competition, firms that share a higher degree of common ownership with their competitors will be less likely to use RPE in executive compensation.*

To test this hypothesis, I obtain detailed data on explicit RPE in executive incentive awards from ISS Incentive Lab.[15] This database contains the compensation items reported in annual proxy statements for the largest 750 public firms in the United States economy. As the set of firms can change from year to year, Incentive Lab back- and forward-fills the data once a firm is ranked within the top 750. Merging with BoardEx, Compustat, CRSP, and Thomson/Refinitiv produces a panel of 11,516 firm-years covering 1375 unique firms from 2010 to 2021, with nearly 1000 firms per year.[16]

Incentive Lab reports compensation awards granted to firms' top executives as disclosed in SEC filings. I focus on performance-based compensation awards (as opposed to unrestricted or time-based grants) where the payout is contingent upon achieving measurable goals, which Incentive Lab classifies as absolute or relative. Payouts from absolute goals are determined by the firm's

[15] To the best of my knowledge, Incentive Lab is the only data provider that tracks RPE awards in executive compensation. Bettis et al. (2014), Bennett et al. (2017), and Bettis et al. (2018) are among the first to study compensation contracts using Incentive Lab data.

[16] The sample period starts in 2010 after the Global Financial Crisis and after BlackRock acquired Barclays Global Investors, which ensures a consistent time series.

performance relative to fixed targets, *e.g.* return-on-equity above 10 %, while payouts from relative goals are measured in comparison to other firms' performance, *e.g.* total shareholder return above peer group median. I define the binary firm-year variable *RPE* as equal to 1 if the firm grants an award with at least one relative goal to their CEO that year and 0 otherwise. Additionally, Incentive Lab reports the fair value of compensation for each award, which I use to build the variable RPE%, defined as the dollar-weighted fraction of total CEO pay tied to relative goals.[17] For firm-years without any RPE, RPE% is zero.

Table 2 reports summary statistics at the firm-year level. The mean of *RPE* is 0.472, meaning that 47 % of firm-years use RPE. There is considerable variation in RPE in both the cross-section and time series. Fig. 1 plots the average values of the variables of interest over the sample period. The figure shows that relative incentives proliferated over the decade from 27 % of sample firms using RPE in 2010 to 62 % in 2021.[18] The mean of *RPE%* is 0.101, implying that conditional on using RPE, about 21 % (=0.101/0.472) of pay is tied to relative awards. This conditional fraction has remained steady over the sample period, perhaps reflecting optimal contracting.

I build measures of common ownership from the Thomson Reuters Institutional Managers (13f) Holdings database, which catalogs the long equity positions of institutional investors with at least $100 million in exchange-traded assets under management. I define *All Direct Ownership* as the total fraction of shares owned by investors in the database. At the mean, these institutions own 72.5 % of shares at sample firms. Next, I define *All Common Ownership* as the value-weighted fraction of firms in the same industry (4-digit *SIC* code) that share at least one institutional blockholder (at least 5 % ownership) in common with the firm. The average firm shares a blockholder with over 54 % of firms in the same industry.

To study how the Big Three asset managers may differentially affect executive incentives, I define *Big 3 Direct Ownership* as the fraction of shares owned by Blackrock, Vanguard, and State Street, and I define *Other Direct Ownership* as the fraction of shares owned by the remaining institutional investors.[19] On average, the Big Three own 16 % of shares in sample firms while the others own 56 %. I define *Big 3 Common Ownership* as the value-weighted fraction of firms in the same industry that share at least one of the Big Three as a blockholder in common with the firm. *Other Common Ownership* is the analogous fraction for same-industry firms commonly held by non-Big Three institutional blockholders. The average firm shares a Big Three blockholder with around 52 % of firms in the same industry, while this fraction is only 11 % for other investors.

Figure 1 shows that at the same time *RPE* rapidly increased, *Big 3 Direct Ownership* rose from 11.5 % to 20 %, and *Big 3 Common Ownership* grew nearly in tandem with *RPE*. In contrast, *Other Direct Ownership* and *Other Common Ownership* declined initially but remained relatively stable over much of the sample period, hovering around 55 % and 10 %, respectively. The persistent and correlated growth of *RPE* and the Big Three's measures challenges the notion that common owners' expanding influence has produced anti-competitive effects. If these asset managers were softening competition, RPE usage would be expected to decline as their ownership increased.

I test Hypothesis 1 by regressing the measures of RPE on the measures of direct and common ownership as in,

$$RPE_{i,t} = \beta_1 Direct\ Ownership_{i,t-1} + \beta_2 Common\ Ownership_{i,t-1} + \lambda\ Controls_{i,t-1} + \eta_i + \eta_t + \varepsilon_{i,t}, \tag{1}$$

where i and t denote firm and year, respectively. The subscripts of '$t-1$' on the ownership measures signify that they are observed at the end of the preceding fiscal year, which is ideal as compensation decisions are usually made in the first quarter of the fiscal year. Controls include measures of industry competitiveness (HHI), the firm's exposure to the industry (stock return correlation with industry), firm performance (stock return and return-on-assets), value and leverage (book-to-market and debt-to-assets), size and complexity (log market capitalization and a multi-segment dummy) and a set of CEO characteristics (CEO-Chair duality, General Ability Index, age, and tenure). All variables are fully defined in Appendix 1. I include firm and year fixed effects to remove unobservable fixed firm-level characteristics and time trends, and I cluster standard errors by firm.

The hypothesis that common owners discourage RPE in executive compensation predicts $\beta_2 < 0$. In contrast, if the institutional investors act as stewards of corporate governance and adhere to standard principal-agent theory, then $\beta_1 > 0$, and if their common ownership does not dilute this preference, then $\beta_2 = 0$. I find no evidence to support Hypothesis 1. When measuring ownership separately for the Big Three *versus* other institutions, I find that higher *Big 3 Direct Ownership* increases RPE use while the measures of common ownership are statistically zero. However, this OLS model is prone to bias if compensation and ownership are endogenously determined. For example, if the Big Three purposely avoid investing in firms that compete with each other, a positive coefficient on *Big 3 Direct Ownership* could follow from standard contract theory and the coefficient on *Big 3 Common Ownership* would not be identified.

To identify the causal effects of the Big Three's ownership on RPE, I exploit the fact that these large asset managers hold a substantial portion of their investments due to passive ownership that tracks major market indices. When firms are added or removed from index lists, the Big Three mechanically increase or decrease their ownership in these firms. Since Incentive Lab covers compensation

[17] Firms frequently distribute multiple awards each year, with considerable variation in the methods of awarding and reporting. For instance, a firm might present a single hybrid award that encompasses both absolute and relative objectives or issue two distinct awards, one for absolute goals and another for relative goals. As Incentive Lab does not report how payouts are split between absolute and relative goals in hybrid awards, I approximate this fraction for each award by dividing the number of relative goals by the total number of goals. Roughly 65 % of awards with RPE are hybrid, and the average hybrid award lists 2.7 absolute goals and 1.4 relative goals. The results are similar if I arbitrarily split hybrid award payouts fifty-to-fifty between absolute and relative goals.

[18] This trend in RPE is consistent with that observed by Bizjak et al. (2022) who also use the Incentive Lab data.

[19] Following Azar et al. (2018), I aggregate fund-level holdings to the institution level.

awards at the 750 largest firms in the U.S. economy, I use membership in the Standard and Poor's 500 (S&P 500) market index to create instruments that identify variation in the Big Three's ownership.[20] In each sample year, around 20 firms are replaced during the S&P 500's reconstitution, generating significant variation in membership; over half of S&P 500 constituents from the start of the sample in 2010 were replaced by the end of the sample in 2021.

My first instrument is a dummy variable, *Firm in S&P 500*, that marks whether the firm is an S&P 500 index constituent. This instrument generates variation in the Big Three's direct ownership in the firm, *Big 3 Direct Ownership*. It is excludable if the firm's choice to use RPE is not a determinant of S&P 500 membership, if an omitted factor does not drive both RPE and membership, and if membership does not influence RPE choice through other pathways. The first two possibilities are remote because the criteria that cause S&P 500 turnover are float-adjusted market capitalization, a float-adjusted stock liquidity ratio, and trailing four-quarter earnings.[21] I address the possibility that performance may cause both RPE choice and index inclusion with control variables. Although the S&P's Index Committee has some discretion over constituent selection, it primarily regards accelerated inclusion of major IPOs, continued inclusion of current constituents following their acquisition by non-constituent firms, and delayed deletion of longtime constituents that might only violate the inclusion criteria temporarily. Additionally, although S&P 500 membership could affect RPE if other investors who prefer it tend to target firms in that index, such activists often act in at least indirect coordination with large passive investors such as the Big Three (Appel et al., 2019). Thus, *Firm in S&P 500* likely satisfies the excludability condition.

My second instrument, *Industry in S&P 500*, is the fraction of other firms in the same industry that are members of the S&P 500. This measure identifies plausibly exogenous variation in the extent of the Big Three's common ownership of the firm and its competitors, *Big 3 Common Ownership*. Since this instrument identifies variation through *other* firms entering or exiting the S&P 500, it is improbable that it could directly affect the firm's governance choices including executive compensation design. Even if a rival's inclusion in the S&P 500 was a factor in the firm's pay-setting process, it would only feasibly affect RPE in whether the firm names the rival as a benchmark peer. That is, even if a firm is differentially inclined to benchmark RPE against rivals in the S&P 500, it is unlikely that it would affect the firm's use of RPE as a component of pay. Thus, *Industry in S&P 500* is also likely excludable.

First-stage regressions confirm these instruments exhibit strong relationships with their respective endogenous variables even in the presence of firm fixed effects. I estimate two-stage least-squares (2SLS) instrumental variables models of Eq. (1). The 2SLS models show that the Big Three increase RPE at the firms they own, and their common ownership does not counteract this effect.

3.2. Say-on-pay votes

My second set of tests investigates 'say-on-pay' advisory shareholder votes on executive compensation. These votes serve as a referendum where shareholders express their approval or disapproval of executive pay packages. Although say-on-pay votes are nonbinding, dissent often produces changes to executive compensation design resulting in increased pay-to-performance sensitivity (Ferri and Maber, 2013; Correa and Lel, 2016).

If common owners aim to soften competition by reducing RPE, I hypothesize that they will use say-on-pay to voice their support for compensation packages that do not contain relative pay elements. Formally:

Hypothesis 2. *If common owners discourage competition, firms that share a higher degree of common ownership with their competitors will receive a higher percentage of votes in favor of compensation in say-on-pay referendums when they incorporate less explicit RPE-based incentive pay into executive compensation contracts.*

To test Hypothesis 2, I obtain firm-level say-on-pay vote tallies from ISS Voting Analytics and define *Votes* as the percentage of votes cast in favor of say-on-pay.[22] As Table 2 reports, the average say-on-pay proposal receives 90 % support. Although most votes receive high support as evidenced by the 25th percentile of 89 %, there is significant variation in the left tail as the standard deviation is over 12 %age points.

I test Hypothesis 2 by regressing *Votes* on the interactions of ownership and the RPE variables as in,

$$Votes_{i,t} = \gamma_1 Direct\ Ownership_{i,t-1} \times RPE_{i,t} + \gamma_2 Common\ Ownership_{t-1} \times RPE_{i,t} + \beta_1 Direct\ Ownership_{i,t-1} + \beta_2 Common\ Ownership_{i,t-1} + \beta_3 RPE_{i,t} + \lambda\ Controls_{i,t-1} + \eta_i + \eta_t + \varepsilon_{i,t}, \tag{2}$$

where subscripts and fixed effects are as before. Controls include those in Eq. (1) and two dimensions of executive compensation that shareholders often consider in say-on-pay votes (*e.g.*, Malenko and Shen, 2016): the level of total CEO compensation (log *Total Compensation*) and the fraction of compensation delivered in stock and option awards (*Pct Stock-based Pay*). In robustness tests, I augment Eq. (2) to include the interactions of the ownership measures with *log Total Compensation* and *Pct Stock-based Pay* as additional controls and find similar results.

The hypothesis that the common owners use say-on-pay to voice their support for pay packages without RPE implies $\gamma_2 < 0$.

[20] Since Incentive Lab requires sample firms to ever appear among the 750 largest in the U.S. economy, I cannot use, for example, Russell 1000 index inclusion as a valid identification strategy.

[21] See S&P Dow Jones Indices: S&P U.S. Indices Methodology (2024) for details on eligibility criteria.

[22] The numerator of *Votes* is always the number of votes in favor. The denominator is the corresponding 'base' in the ISS data, which can be votes in favor plus votes against, votes in favor plus votes against plus abstentions, or the number of outstanding shares.

Table 3
RPE Peer selection summary statistics.

Panel A: Summary statistics by firm-peer-year

	mean	sd	p25	p50	p75
RPE Peer	0.065	0.247	0.000	0.000	0.000
Big 3 Direct Ownership	0.187	0.075	0.152	0.195	0.235
Big 3 Common Block	0.743	0.437	0.000	1.000	1.000
Other Direct Ownership	0.536	0.182	0.432	0.554	0.648
Other Common Block	0.081	0.273	0.000	0.000	0.000
Product Similarity	0.362	0.331	0.137	0.401	0.624
Return Correlation	0.054	0.063	0.004	0.033	0.080
Similar Stock Return	0.357	0.479	0.000	0.000	1.000
Similar Return-on-Assets	0.502	0.500	0.000	1.000	1.000
Similar Book-to-Market	0.319	0.466	0.000	0.000	1.000
Similar Leverage	0.356	0.479	0.000	0.000	1.000
Similar MCAP	0.239	0.427	0.000	0.000	0.000
Both Single- or Multi-segment	0.727	0.445	0.000	1.000	1.000
Same 1-digit SIC industry	1.000	0.000	1.000	1.000	1.000
Same 2-digit SIC industry	0.290	0.454	0.000	0.000	1.000
Same 3-digit SIC industry	0.167	0.373	0.000	0.000	0.000
Same 4-digit SIC industry	0.138	0.345	0.000	0.000	0.000
Firm-Peer-Year Observations	527,283				

Panel B: Firm-year average differences between peers and non-peers

	Peers	Non-Peers	Difference
RPE Peer	1.000	0.000	1.000***
Big 3 Direct Ownership	0.183	0.182	0.000
Big 3 Common Block	0.755	0.714	0.041***
Other Direct Ownership	0.538	0.539	−0.001
Other Common Block	0.134	0.079	0.055***
Product Similarity	0.555	0.341	0.214***
Return Correlation	0.144	0.051	0.093***
Similar Stock Return	0.444	0.330	0.114***
Similar Return-on-Assets	0.559	0.451	0.108***
Similar Book-to-Market	0.413	0.311	0.102***
Similar Leverage	0.519	0.344	0.176***
Similar MCAP	0.404	0.223	0.181***
Both Single- or Multi-segment	0.758	0.737	0.022***
Same 1-digit SIC industry	1.000	1.000	0.000
Same 2-digit SIC industry	0.827	0.291	0.537***
Same 3-digit SIC industry	0.643	0.143	0.500***
Same 4-digit SIC industry	0.565	0.108	0.457***
Number of Peers	11.274	160.146	
Peers added this year	1.238		
Peers removed next year	0.854		
Firm-Year Observations	3047		

This table presents summary statistics at the firm-peer-year level. The sample consists of all firms in the Incentive Lab database from 2010 to 2021 with non-missing data items after merging with BoardEx, Compustat, CRSP, and Thomson/Refinitiv that use RPE and benchmark against a peer group, or 3047 firm-years. Counterfactual peers are firms in the same 1-digit *SIC* industry, producing 527,283 firm-peer-year observations. Variables are defined in Appendix 1. Continuous variables are winsorized at the 1st and 99th levels.

However, as discussed above regarding Appendix 2, the stewardship-focused institutional investors such as the Big Three may prefer RPE in executive compensation because it tethers pay to performance while limiting 'pay-for-luck'. If these investors prefer RPE, then $\gamma_1 > 0$, and if common ownership is not a mitigating factor, then $\gamma_2 = 0$. My results are consistent with the latter.

3.3. Peer selection

My third set of tests investigate the possibility that the common owners circumvent RPE's competitive effects by altering firms' RPE benchmark groups. Each RPE goal lists the underlying benchmark group used as the basis of comparison when evaluating relative performance. Using Incentive Lab, I classify these groups into three categories: a peer group (*e.g.* panel C of Table 1), an industry-specific index (*e.g.* the Dow Jones U.S. Banks Index), or a broad market index (*e.g.* the Russell 1000). If the common owners wish to discourage competition while using RPE in compensation, they may choose to benchmark against a market index that is too broad to capture gains from industry-level collusion or against a custom peer group that is designed to exclude commonly held competitors. That is, they will be less inclined to benchmark against an industry-specific index because that would incentivize managers to compete more fiercely against firms in their own industries.

I test this in two stages, first by studying benchmark group choice, and second by studying individual peer firm selection. I formalize the first hypothesis as follows:

Hypothesis 3a. *If common owners discourage competition, firms that share a higher degree of common ownership with their competitors will be more likely to benchmark RPE against a peer group or market index and less likely to benchmark against an industry index.*

To test Hypothesis 3a, I define three binary variables, *Relative to Peer Group*, *Relative to Industry Index*, and *Relative to Market Index*, that indicate whether the firm benchmarks RPE against each corresponding group. Table 2 reports that 55 % of firms that use RPE benchmark performance against a peer group, 18 % benchmark against an industry index, and 34 % benchmark against a broad market index.[23]

I estimate Eq. (1) with these three measures as dependent variables. If common owners use RPE in a manner that softens competition, then the β-coefficients on ownership and common ownership measures will be positive for the *Peer Group* or *Market Index* outcomes, and negative for the *Industry Index* outcome. However, I find no evidence of this.

Second, I study whether the common owners tailor firms' custom RPE peer groups to avoid pitting their portfolio constituents against each other. The idea is that even if common owners encourage the use of RPE, they might pressure firms to remove commonly owned rivals from their peer groups to limit competition. Formally:

Hypothesis 3b. *If common owners discourage competition, firms will be less likely to include rivals in their RPE peer groups when they share a common owner.*

To test Hypothesis 3b, I construct a dataset of firm-peer pairs and define the binary variable *RPE Peer* as equal to 1 if the benchmarking firm names the other in its RPE peer group that year and 0 if the firm is not named. For practicality and tractability, I restrict the counterfactual peers to those in the same 1-digit *SIC* code as the benchmarking firm since firms predominately select RPE peers from their industry (*e.g.* Gong et al., 2011).

Table 3 reports summary statistics at the firm-peer-year level. The average peer group contains 11.274 firms. While peer groups are fairly consistent from year to year, they are not static. On average, a firm adds 1.238 new peer firms to the group each year and removes 0.854 firms when moving into the next year. These numbers imply an addition rate of 11 % (=1.238/11.274) and a deletion rate of 7.6 % (=0.854/11.274). This dynamic suggests that adjustments to RPE peer groups, though modest, occur regularly and could provide a strategic mechanism for influential stakeholders to drive change. An influential common owner might exert pressure on firms to drop a commonly held firm from the RPE peer group, particularly if the two firms are fierce competitors. For instance, as Feichter et al. (2022) note, the Big Three may influence changes in their portfolio firms through "hidden power" *via* private engagements with management.

Moreover, narrowly adjusting the peer group is arguably more feasible than attempting a full redesign of a compensation program to exclude RPE. Compensation programs typically consist of finely tuned short-term and long-term goals, require board ratification, and are subject to an advisory shareholder vote. The fixed costs associated with a comprehensive redesign can be substantial, including consulting fees, legal and compliance-related expenses, and the costs of stakeholder engagement. Thus, excluding a commonly held competitor from the RPE group is potentially a cost-effective path of least resistance to discourage competition.

To test whether common ownership affects peer selection, I define pairwise dummy variables, *Big 3 Common Block* and *Other Common Block*, at the firm-peer-year level as equal to 1 an investor simultaneously holds blocks in both the firm and its potential peer. Constraining potential peers to the same 1-digit *SIC* code, about 160 potential candidates are not selected, producing a baseline probability that a firm chooses a given peer of 6.5 % (=11.274/160.146). Panel A of Table 3 reports that 74.3 % of sample pairs share a Big Three blockholder. Panel B shows that the fraction is 75.5 % among RPE peers and only 71.4 % for non-RPE peers. For non-Big Three institutions, 8.1 % share a common blockholder but this rises to 13.4 % for RPE peers that are selected and drops to 7.9 % for non-peers. These patterns are the opposite of what Hypothesis 3b suggests.

To further test whether common owners discourage RPE between close competitors, I adopt two measures of pairwise competitiveness. *Product Similarity* is the pairwise product market similarity score based on textual analysis of product descriptions by Hoberg and Phillips (2010, 2016). *Return Correlation* is pairwise stock return correlation based on the prior year's daily returns, which provides a market-based measure of competitiveness.

I test whether common ownership influences RPE peer selection by estimating models of the form,

$$\begin{aligned} RPE\ Peer_{i,j,t} = {} & \phi_1 Direct\ Ownership_{i,t-1} \times Competition_{i,j,t-1} + \phi_2 Common\ Block_{i,j,t-1} \times Competition_{i,j,t-1} + \phi_3 Common\ Block_{i,j,t-1} \\ & + \phi_4 Competition_{i,j,t-1} + \lambda\ Controls_{i,j,t-1} + \eta_{i,j} + \eta_{i,t} + \eta_{j,t} + \varepsilon_{i,j,t}, \end{aligned} \tag{3}$$

for firm i and potential peer j in year t. *Controls* is a collection of firm-peer-year measures that capture other dimensions of similarity including a set of dummies that, in an approach akin to Faulkender and Yang (2013), indicate whether the firms are within one sample standard deviation of each other in terms of firm-level observables. Granular firm × peer fixed effects absorb unobserved pairwise traits, while firm × year and peer × year fixed effects account for within-firm-year and within-peer-year heterogeneity. The direct effects of *Direct Ownership* and *Other Direct Ownership* are absorbed by the firm × year fixed effects. I cluster standard errors at the firm-pair level.

The hypothesis that the common owners discourage competition among their portfolio constituents by not rewarding them for outperforming each other predicts $\phi_2 < 0$ and $\phi_3 < 0$. On the other hand, if the stewardship-focused investors use RPE to filter out

[23] These fractions sum to over 100 % because firms sometimes use multiple benchmarks.

Table 4
RPE and common ownership.

Panel A: Binary RPE Outcome

	(1)	(2)	(3)	(4)	(5)
	RPE	RPE	RPE	RPE	RPE
Institutional Direct Ownership	−0.014				
	(−0.33)				
Institutional Common Ownership	0.180***				
	(6.24)				
Big 3 Direct Ownership		1.305***		1.591***	0.296**
		(7.82)		(9.10)	(1.99)
Big 3 Common Ownership		0.028		0.043	−0.008
		(0.96)		(1.48)	(−0.37)
Other Direct Ownership			−0.005	−0.265***	−0.051
			(−0.10)	(−5.33)	(−1.19)
Other Common Ownership			0.020	0.014	0.024
			(0.52)	(0.37)	(0.95)
Controls	Yes	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	11,516	11,516	11,516	11,516	11,426
Adjusted R-squared	0.146	0.166	0.132	0.173	0.625

Panel B: Continuous RPE Outcome

	(1)	(2)	(3)	(4)	(5)
	RPE%	RPE%	RPE%	RPE%	RPE%
Institutional Direct Ownership	0.004				
	(0.30)				
Institutional Common Ownership	0.033***				
	(3.98)				
Big 3 Direct Ownership		0.309***		0.365***	0.116***
		(5.39)		(5.95)	(2.60)
Big 3 Common Ownership		−0.000		0.004	−0.004
		(−0.05)		(0.40)	(−0.62)
Other Direct Ownership			0.005	−0.052***	−0.025*
			(0.33)	(−3.29)	(−1.88)
Other Common Ownership			−0.006	−0.006	0.008
			(−0.59)	(−0.58)	(1.04)
Controls	Yes	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	11,516	11,516	11,516	11,516	11,426
Adjusted R-squared	0.077	0.090	0.071	0.093	0.576

This table shows that ownership by the Big Three asset management firms is positively associated with RPE in executive compensation, and this effect is not mitigated by common ownership. *RPE* is a dummy variable equal to one if the firm uses RPE in CEO compensation. *RPE%* is the dollar fraction of CEO compensation tied to RPE awards. *Institutional Direct Ownership* is the fraction of the firm's stock owned by institutional investors. *Institutional Common Ownership* is the fraction of same-industry firms that share at least one institutional investor as a blockholder with the firm. *Big 3 Direct Ownership* is the fraction of shares owned by BlackRock, State Street, and Vanguard. *Big 3 Common Ownership* is the fraction of same-industry firms that share a common Big Three blockholder with the firm. *Other Direct Ownership* is the fraction of the firm's stock owned by institutional investors excluding the Big Three. *Other Common Ownership* is the fraction of same-industry firms that share at least one non-Big Three institutional investor as a blockholder with the firm. Variables are defined in Appendix 1. Controls are suppressed for brevity but are reported in the Internet Appendix. Estimation is OLS with fixed effects. Standard errors are clustered by firm; t-statistics are in parenthesis. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

shared exogenous shocks as contract theory suggests, then $\phi_1 > 0$ and $\phi_2 = 0$, which is precisely what I find.

4. Results

4.1. RPE use

Table 4 reports results from regressing *RPE* and *RPE%* on the measures of direct and common ownership as in Eq. (1). All specifications include year-level fixed effects, and the final column includes firm-level fixed effects.

Column (1) reports the estimated relationship between RPE and the aggregate measures of institutional ownership. The effect of *Institutional Direct Ownership* is small and statistically insignificant, whereas the effect of *Institutional Common Ownership* is large, positive, and statistically significant. These findings are inconsistent with Hypothesis 1, but it remains unclear why common ownership

Table 5
RPE and big three common ownership, instrumental variables.

	First Stage	First Stage	First Stage	First Stage	2SLS	2SLS	2SLS	2SLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Big 3 Direct Ownership	Big 3 Direct Ownership	Big 3 Common Ownership	Big 3 Common Ownership	RPE	RPE	RPE%	RPE%
Big 3 Direct Ownership					3.281***	3.075*	0.628**	0.527
					(3.68)	(1.69)	(2.31)	(0.98)
Big 3 Common Ownership					0.049	0.169	0.012	0.097
					(0.21)	(0.71)	(0.17)	(1.37)
Firm in S&P 500	0.047***	0.015***	0.111***	0.003				
	(11.51)	(4.48)	(5.71)	(0.19)				
Industry in S&P 500	0.010*	0.003	0.175***	0.217***				
	(1.86)	(0.53)	(6.53)	(5.42)				
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	No	Yes	No	Yes	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	11,516	11,426	11,516	11,426	11,516	11,426	11,516	11,426
Adjusted R-squared	0.261	0.760	0.140	0.621	0.051	−0.162	0.037	−0.112
Kleibergen-Paap F-statistic					26.969	10.291	26.969	10.291
Stock-Yogo 10 % critical value					7.03	7.03	7.03	7.03

This table reports instrumental variables regressions for the effect of Big Three ownership and common ownership on RPE use. A dummy variable that indicates the firm is in the S&P 500 instruments for *Big 3 Direct Ownership*, which is the fraction of shares owned by BlackRock, State Street, and Vanguard. The fraction of same-industry firms in the S&P 500 instruments for *Big 3 Common Ownership*, which is the fraction of same-industry firms that share a Big Three blockholder with the firm. *RPE* is a dummy variable equal to one if the firm uses RPE in CEO compensation. *RPE%* is the dollar fraction of CEO compensation tied to RPE awards. Variables are defined in Appendix 1. Controls are suppressed for brevity but are reported in the Internet Appendix. Standard errors are clustered by firm; t-statistics are in parenthesis. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

would increase RPE.

Separating institutional investors into the Big Three *versus* all others reveals that the overall effect of common ownership is driven by the direct ownership of the Big Three. The β_1 coefficient on *Big 3 Direct Ownership* in Panel A column (2) is 1.305, which implies that a one standard deviation increase in their ownership corresponds to a 20 % increase (=1.305 × 0.073/0.472) in the unconditional probability that the firm uses RPE, relative to the sample mean.[24] In contrast, the coefficients on *Big 3 Common Ownership* are small and statistically indistinguishable from zero in all columns. The point estimate and standard error for β_2 are 0.028 and 0.029, respectively, which implies that the one-sided 95 % confidence lower bound on the point estimate is −0.020 (=0.028–1.645 × 0.029). A coefficient of −0.020 predicts that a one standard deviation increase in *Big 3 Common Ownership* corresponds to a relatively small 1.55 % reduction (= − 0.020 × 0.363/0.472) in RPE use relative to the sample mean. Thus, we can say with 95 % confidence that the true effect of a one-sigma increase in *Big 3 Common Ownership* on RPE is no more negative than a 1.55 % decline; the true effect is likely zero or even positive. Based on these estimates, the Big Three's common ownership does not reduce firms' RPE use.

Column (3) reports the effects of other institutional investors' direct and common ownership on RPE. Both effects are small and tightly within the neighborhood of zero. Focusing on the effect of *Other Common Ownership*, the estimate for β_2 is 0.020 with a standard error of 0.039. This implies a one-sided 95 % confidence lower bound of −0.043 (=0.020–1.645 × 0.039). Translating, the impact of a one-sigma increase in *Other Common Ownership* is unlikely to lower RPE by more than a modest 1.76 % decline (= − 0.043 × 0.192/0.472).

Columns (4) and (5) report results when both Big Three and other institutions' direct and common ownership are included as separate variables. The effect of *Big 3 Direct Ownership* on RPE is strongly positive and persists even in the presence of firm-level fixed effects. In contrast, the effect of *Other Direct Ownership* on RPE is negative, although only weakly so when within-firm variation is removed. This result is intuitive because non-Big Three institutions are smaller and potentially less influential or less interested in influencing governance stewardship than the Big Three.[25] The effects of *Big 3 Common Ownership* and *Other Common Ownership* remain near zero in all specifications.

To investigate whether multicollinearity between the measures of *Direct Ownership* and *Common Ownership* complicates this

[24] All in-text calculations are performed using full precision but are rounded for display, which may cause visual rounding errors.

[25] Affecting changes within firms is costly, and the Big Three achieve economies of scale that other institutional investors cannot. Additionally, other investors' objective functions may be different than the Big Three, who the results show favor RPE. For example, other institutional investors might hold shares primarily to gain exposure to a firm's industry and may prefer compensating managers based on overall performance rather than relative performance. Moreover, RPE is a relatively modern compensation apparatus (recall Fig. 1) and other institutional investors may lack the incentive or initiative to advocate for its adoption.

inference, I examine variance inflation factors (VIF) for the regression models. The VIF factor values are reported in Appendix 3. In column (1), the VIF values are 2.00 for *Big 3 Ownership* and 1.77 for *Big 3 Common Ownership*. These values are well below the commonly accepted threshold of 10, which is often used to signal severe multicollinearity issues. Moving to the most saturated model in column (6), the VIF values on the coefficients of interest are 6.62 for *Big 3 Direct Ownership*, 3.70 for *Big 3 Common Ownership*, 4.42 for *Other Direct Ownership*, and 1.85 for *Other Common Ownership*. These values suggest that the variables of interest have sufficient variance for both between- and within-estimators. All VIF values on the parameters of interest fall within the range considered to indicate a low to moderate degree of multicollinearity, suggesting that the relationships between these regressors (and the control variables) do not unduly inflate the standard errors of the estimated coefficients.

Table 5 presents results from instrumental variables specifications of eq. (1), where *Firm in S&P 500* instruments for *Big 3 Direct Ownership* and *Industry in S&P 500* instruments for *Big 3 Common Ownership*. The first four columns report first-stage regressions, confirming strong relationships between the instruments and the respective endogenous regressors. The relevant t-statistics on *Firm in S&P 500* are 11.51 and 4.48 with and without firm-level fixed effects, and the corresponding t-statistics for *Industry in S&P 500* are 6.53 and 5.42. This high level of significance is important because it facilitates accurate estimation of standard errors in the second stage. This, in turn, allows me to distinguish between effects that are only statistically insignificant due to lack of power and those that are truly zero.

These first-stage relationships are also economically significant. The coefficient on *Firm in S&P 500* in column (1) is 0.047, implying that all else equal, firms in the S&P 500 have 4.7 percentage points higher *Big 3 Direct Ownership* than those not in the index. This coefficient is 0.015 in column (2) with the addition of firm fixed effects, meaning that *Big 3 Direct Ownership* increases (or decreases) by 1.5 percentage points when a firm is added to (or removed from) the S&P 500. Columns (3) and (4) show that *Industry in S&P 500* strongly predicts *Big 3 Common Ownership*. The point estimate of 0.217 in column (4) implies that a one-sigma increase in the instrument produces a 13 % increase (=0.217 × 0.291/0.486) in *Big 3 Common Ownership,* relative to the sample mean.

To formally assess the relevance condition, I report Kleibergen-Paap rank Wald F-statistics and the associated benchmark critical value in the last two rows of Table 5. The F-statistic is 26.969 in the specifications without firm-level fixed effects and 10.291 in the specifications with firm-level fixed effects. In both models, the KP F-statistic exceeds the Stock-Yogo critical value of 7.03 (the 10 % maximal IV bias threshold for a model with two endogenous regressors and two excluded instruments), indicating that the instruments are sufficiently strong. Therefore, I reject the null hypothesis of weak instruments. Thus, in addition to being excludable (recall section 3.1), the instruments satisfy the relevance condition.

The last four columns of Table 5 report 2SLS results. The effect of *Big 3 Direct Ownership* on RPE is statistically positive in columns (5) thru (7) and is negative in none. The magnitude of this effect increases relative to OLS and is economically significant yet plausible. The instrumented coefficient on *Big 3 Direct Ownership* in the presence of firm-level fixed effects in column (6) is 3.075, implying that the 1.5 percentage point higher *Big 3 Direct Ownership* associated with the firm joining the S&P 500 in column (2) increases the probability that the firm uses RPE by 4.6 percentage points or about 9.8 % of the unconditional sample mean (=3.075 × 0.015/0.470). Extrapolating, the full effect of a one-sigma increase in *Big 3 Direct Ownership*, or 7.2 % of a firm's shares, causes a 47.1 % increase (=3.075 × 0.072/0.470) in the probability that the firm uses RPE, relative to the sample mean. This magnitude is large but feasible considering the influence that 7.2 percent of a company's shares can have on corporate policy.

In contrast, the instrumented effect of *Big 3 Common Ownership* on RPE use remains small and within the statistical neighborhood of zero, meaning that their common ownership likely has no effect on firms' RPE use. In column (6), the instrumented coefficient on *Big 3 Common Ownership* in the presence of firm fixed effects is 0.169 with a standard error of 0.238, meaning that the one-sided 95 % confidence lower bound on this estimate is −0.223 (=0.169–1.645 × 0.238). Therefore, with 95 % confidence, the true reduction in RPE due to a one standard deviation increase in *Big 3 Common Ownership* is no more than a 17.7 % decline in probability relative to the sample mean (= − 0.223 × 0.373/0.470). Even at the 95 % lower bound, the implied decline in *RPE* due to a one-sigma increase in *Big 3 Common Ownership* is considerably smaller than the 47.1 % increase implied by the baseline point estimate for the effect of a one-sigma increase in *Big 3 Direct Ownership*. The true effect of *Big 3 Common Ownership* on *RPE* is likely zero or even positive. Taken together, firms with more shares owned by the Big Three asset managers are more likely to use RPE in executive compensation, and this propensity is not offset by the Big Three's common ownership of rival firms.

4.2. Say-on-pay votes

Table 6 reports results from regressing say-on-pay vote outcomes on measures of the Big Three's ownership and common ownership as in Eq. (2). The coefficients on the interactions of *Big 3 Direct Ownership* and the RPE measures are statistically positive in all columns while the interactions involving *Big 3 Common Ownership*, *Other Direct Ownership*, and *Other Common Ownership* are statistically significant in none.

Focusing on Panel A, The coefficient on *Big 3 Direct Ownership* × *RPE* in the presence of firm-level fixed effects, in column (2), is 0.198. A one standard deviation increase in *Big 3 Direct Ownership* at a firm that uses RPE is expected to produce 1.4 percentage points more support (=0.198 × 0.072) for say-on-pay than a firm that does not. Relative to *Votes* standard deviation, this represents an

Table 6
RPE, common ownership, and Say-on-Pay.

Panel A: Binary RPE Interactions

	(1)	(2)	(3)	(4)	(5)	(6)
	Votes	Votes	Votes	Votes	Votes	Votes
Big 3 Direct Ownership × RPE	0.149**	0.198***			0.160**	0.232***
	(2.28)	(2.92)			(2.26)	(3.22)
Big 3 Common Ownership × RPE	−0.004	−0.007			−0.001	0.003
	(−0.36)	(−0.59)			(−0.11)	(0.20)
Other Direct Ownership × RPE			−0.001	0.006	−0.015	−0.012
			(−0.06)	(0.26)	(−0.79)	(−0.56)
Other Common Ownership × RPE			0.037	0.002	−0.016	−0.114
			(0.64)	(0.03)	(−0.24)	(−1.49)
Big 3 Direct Ownership	−0.084	−0.092			−0.079	−0.108
	(−1.49)	(−1.51)			(−1.29)	(−1.58)
Big 3 Common Ownership	0.009	−0.004			0.009	−0.006
	(1.02)	(−0.42)			(0.93)	(−0.61)
Other Direct Ownership			−0.010	−0.003	−0.003	0.006
			(−0.64)	(−0.14)	(−0.17)	(0.30)
Other Common Ownership			0.005	−0.048	−0.000	0.019
			(0.11)	(−0.72)	(−0.01)	(0.25)
RPE	−0.006	−0.019*	0.012	0.007	0.001	−0.007
	(−0.67)	(−1.96)	(1.13)	(0.56)	(0.07)	(−0.59)
log Total Compensation	−0.034***	−0.006	−0.034***	−0.006	−0.034***	−0.006
	(−7.64)	(−1.42)	(−7.49)	(−1.45)	(−7.72)	(−1.41)
Pct Stock-based Pay	0.022**	0.027**	0.022*	0.027**	0.023**	0.027**
	(2.02)	(2.43)	(1.95)	(2.38)	(2.14)	(2.44)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	No	Yes	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9309	9230	9309	9230	9309	9230
Adjusted R-squared	0.061	0.282	0.060	0.280	0.061	0.282

Panel B: Continuous RPE Interactions

	(1)	(2)	(3)	(4)	(5)	(6)
	Votes	Votes	Votes	Votes	Votes	Votes
Big 3 Direct Ownership × RPE%	0.359*	0.351*			0.353*	0.388*
	(1.93)	(1.65)			(1.68)	(1.68)
Big 3 Common Ownership × RPE%	0.010	−0.019			0.018	0.000
	(0.25)	(−0.43)			(0.44)	(0.00)
Other Direct Ownership × RPE%			0.043	0.019	0.019	−0.007
			(0.64)	(0.25)	(0.28)	(−0.09)
Other Common Ownership × RPE%			0.121	0.016	−0.105	−0.200
			(0.57)	(0.07)	(−0.41)	(−0.67)
Big 3 Direct Ownership	−0.023	−0.028			−0.024	−0.018
	(−0.78)	(−0.79)			(−0.48)	(−0.28)
Big 3 Common Ownership	−0.039	−0.022			0.006	−0.006
	(−0.84)	(−0.41)			(0.76)	(−0.69)
Other Direct Ownership			−0.017	−0.002	−0.015	−0.000
			(−1.18)	(−0.12)	(−1.01)	(−0.02)
Other Common Ownership			0.021	−0.048	0.005	−0.017
			(0.47)	(−0.77)	(0.11)	(−0.25)
RPE%	0.006	−0.006	0.006	0.013	−0.024	−0.013
	(0.76)	(−0.75)	(0.16)	(0.30)	(−0.58)	(−0.28)
log Total Compensation	−0.032***	−0.005	−0.033***	−0.005	−0.033***	−0.005
	(−7.21)	(−1.26)	(−7.19)	(−1.28)	(−7.23)	(−1.28)
Pct Stock-based Pay	0.018	0.026**	0.019*	0.026**	0.020*	0.026**
	(1.64)	(2.30)	(1.67)	(2.28)	(1.76)	(2.30)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	No	Yes	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9309	9230	9309	9230	9309	9230
Adjusted R-squared	0.060	0.280	0.059	0.280	0.060	0.280

This table shows that firms with higher Big Three ownership receive more support in advisory say-on-pay votes if they use RPE in CEO compensation, and this effect is not mitigated by common ownership. *Votes* is the fraction of shareholder votes in favor of the compensation program. *RPE* is a dummy variable equal to one if the firm uses RPE in CEO compensation. *RPE%* is the dollar fraction of CEO compensation tied to RPE awards. *Big 3 Direct Ownership* is the fraction of shares owned by BlackRock, State Street, and Vanguard. *Big 3 Common Ownership* is the fraction of same-industry firms that share a common Big Three blockholder with the firm. *Other Direct Ownership* is the fraction of the firm's stock owned by institutional

investors excluding the Big Three. *Other Common Ownership* is the fraction of same-industry firms that share at least one non-Big Three institutional investor as a blockholder with the firm. Variables are defined in Appendix 1. Controls are suppressed for brevity but are reported in the Internet Appendix. Estimation is OLS with fixed effects. Standard errors are clustered by firm; t-statistics are in parenthesis. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7
RPE benchmark choice and common ownership.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Peer Group	Peer Group	Peer Group	Industry Index	Industry Index	Industry Index	Market Index	Market Index	Market Index
Big 3 Direct Ownership	0.105		0.178	−0.007		0.112	0.088		0.097
	(0.53)		(0.82)	(−0.04)		(0.51)	(0.45)		(0.47)
Big 3 Common Ownership	0.004		0.008	0.004		0.016	−0.026		−0.029
	(0.09)		(0.21)	(0.13)		(0.48)	(−0.68)		(−0.75)
Other Direct Ownership		−0.045	−0.079		−0.102*	−0.131*		−0.012	−0.007
		(−0.89)	(−1.30)		(−1.66)	(−1.90)		(−0.25)	(−0.14)
Other Common Ownership		0.023	0.024		−0.016	−0.015		0.050	0.052
		(0.61)	(0.66)		(−0.43)	(−0.41)		(1.53)	(1.60)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5326	5326	5326	5326	5326	5326	5326	5326	5326
Adjusted R-squared	0.704	0.704	0.704	0.645	0.645	0.645	0.687	0.688	0.688

This table studies the choice of RPE benchmark reference groups as a function of common ownership. The outcome indicates whether performance is benchmarked against a *Peer Group*, an *Industry Index*, or a *Market Index*. *Big 3 Direct Ownership* is the fraction of shares owned by BlackRock, State Street, and Vanguard. *Big 3 Common Ownership* is the fraction of same-industry firms that share a common Big Three blockholder with the firm. *Other Direct Ownership* is the fraction of the firm's stock owned by institutional investors excluding the Big Three. *Other Common Ownership* is the fraction of same-industry firms that share at least one non-Big Three institutional investor as a blockholder with the firm. Variables are defined in Appendix 1. Controls are suppressed for brevity but are reported in the Internet Appendix. Estimation is OLS with fixed effects. Standard errors are clustered by firm; t-statistics are in parenthesis. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

increase of 11.7 % (=0.198 × 0.072/0.122).[26] This effect's statistical and economic significance suggests that the Big Three support RPE in executive compensation despite its competitive effects.

In contrast, the coefficients on *Big 3 Common Ownership* and the RPE measures are small and insignificant in all columns. In column (2), the point estimate is −0.007 and its standard error is 0.013. The one-sided 95 % confidence lower bound for the coefficient is −0.028 (= − 0.007–1.645 × 0.013). At this lower bound, a one standard deviation increase in *Big 3 Common Ownership* corresponds to a 1.0 percentage point decrease in support (−0.028 × 0.373). Thus, even at the 95 % lower bound, the negative effect of *Big 3 Common Ownership* × *RPE* is over a magnitude smaller than the 11.7 % positive effect implied by the point value estimate on *Big 3 Direct Ownership* × *RPE*. The true effect of this interaction is likely zero.

The coefficients on *Other Direct Ownership* × *RPE* and *Other Common Ownership* × *RPE* are insignificant in all specifications. Thus, although the effect of *Other Direct Ownership* on RPE use is negative in some of the Table 4 specifications, these institutions do not vocalize any preference regarding RPE with their voting behavior. This is consistent with other institutions being neutral or perhaps apathetic to RPE in executive compensation. In Column (4) of the present table, the point estimate and standard error for *Other Common Ownership*× *RPE* are 0.002 and 0.066, respectively. This implies a lower-95 % confidence bound of −0.107 (=0.002–1.645 × 0.066), suggesting that a one-sigma increase in *Other Common Ownership* would reduce *Votes* by a modest 2 percentage points. This lower bound is again dwarfed in comparison to the effect of *Big 3 Direct Ownership*× *RPE*, and the true effect *Other Common Ownership*× *RPE* is likely zero.

Additionally, I address the possibility that the RPE measures capture an omitted dimension of executive compensation that correlates with RPE. For example, if the institutions vote against overpaying executives and if firms that overpay their executives typically do not use RPE, this omitted term may confound the results. I augment Eq. (2) by interacting the measures of direct and common ownership with log *Total Compensation* and *Pct Stock-based Pay* and re-estimate. The resulting table, which contains twelve interaction terms, has been relegated to the Internet Appendix due to its length. However, the inference is unchanged: the interactions of *Big 3 Direct Ownership* and the RPE measures are statistically positive and the interactions of the common ownership and RPE measures are

[26] To understand the effect of initiating RPE on say-on-pay support, one must interpret the interaction effect in conjunction with the direct effect of RPE. A firm that uses RPE and is at the 25th percentile of *Big 3 Direct Ownership* receives 0.5 percentage points more support in say-on-pay than if it did not use RPE (=0.198 × 0.121–0.019), while a firm at the 75th percentile of *Big 3 Direct Ownership* receives 2.2 percentage points more support (=0.198 × 0.208–0.019).

Table 8
RPE peer selection and common ownership.

Panel A: Product Similarity

	(1)	(2)	(3)	(4)	(5)	(6)
	RPE Peer	RPE Peer	RPE Peer	RPE Peer	RPE Peer	RPE Peer
Big 3 Direct Ownership × Similarity		0.517***				0.622***
		(2.64)				(2.89)
Big 3 Common Block × Similarity		0.031				−0.003
		(1.42)				(−0.13)
Other Direct Ownership × Similarity				0.050		−0.024
				(0.62)		(−0.27)
Other Common Block × Similarity				0.029		0.030
				(1.05)		(1.10)
Big 3 Direct Ownership	*Absorbed*	*Absorbed*			*Absorbed*	*Absorbed*
Big 3 Common Block	0.003*	0.002			0.003	0.003*
	(1.82)	(0.89)			(1.63)	(1.73)
Other Direct Ownership			*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*
Other Common Block			0.001	−0.001	0.001	−0.001
			(0.43)	(−0.99)	(0.42)	(−1.02)
Similarity	0.112***	−0.010	0.112***	0.036	0.112***	−0.036
	(5.42)	(−0.25)	(5.42)	(0.78)	(5.42)	(−0.74)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm × Peer FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Peer × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	484,720	484,720	484,720	484,720	484,720	484,720
Adjusted R-squared	0.830	0.830	0.830	0.830	0.830	0.830

Panel B: Return Correlation

	(1)	(2)	(3)	(4)	(5)	(6)
	RPE Peer	RPE Peer	RPE Peer	RPE Peer	RPE Peer	RPE Peer
Big 3 Direct Ownership × Correlation		0.041***				0.048***
		(3.55)				(3.91)
Big 3 Common Ownership × Correlation		0.000				−0.000
		(0.10)				(−0.13)
Other Direct Ownership × Correlation				−0.006		−0.015***
				(−1.22)		(−2.62)
Other Common Ownership × Correlation				−0.000		0.000
				(−0.01)		(0.06)
Big 3 Direct Ownership	*Absorbed*	*Absorbed*			*Absorbed*	*Absorbed*
Big 3 Common Ownership	0.003*	0.003*			0.003	0.003*
	(1.82)	(1.74)			(1.64)	(1.67)
Other Direct Ownership			*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*
Other Common Ownership			0.001	0.001	0.001	0.000
			(0.45)	(0.46)	(0.44)	(0.40)
Correlation	0.005***	−0.003	0.005***	0.009***	0.005***	0.004
	(6.52)	(−1.28)	(6.51)	(3.31)	(6.52)	(1.32)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm × Peer FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Peer × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	484,720	484,720	484,720	484,720	484,720	484,720
Adjusted R-squared	0.830	0.830	0.830	0.830	0.830	0.830

This table shows that firms with higher Big Three ownership are more likely to choose close competitors as RPE peers, and this is not lessened by pairwise common ownership. *RPE Peer* is a dummy variable that indicates if the firm was chosen as an RPE peer. *Big 3 Direct Ownership* is the fraction of shares owned by BlackRock, State Street, and Vanguard. *Big 3 Common Block* is a dummy variable that equals one if the firm-pair share a Big Three blockholder. *Other Direct Ownership* is the fraction of the firm's stock owned by institutional investors excluding the Big Three. *Other Common Block* is a dummy variable that equals one if the firm-pair share a non-Big Three blockholder. *Similarity* is the Hoberg-Phillips pairwise text-based similarity score of product market overlap. *Correlation* is pairwise stock return correlation. Variables are defined in Appendix 1. Controls are suppressed for brevity but are reported in the Internet Appendix. Estimation is OLS with fixed effects. Standard errors are clustered by firm-pair; t-statistics are in parenthesis. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

zero. I find no evidence that institutional investors use say-on-pay to voice their support for compensation that excludes relative incentives.

4.3. Peer selection

Table 7 presents results from regressing *Relative to Peer Group*, *Relative to Industry Index*, and *Relative to Market Index* on the measures of institutions' direct and common ownership, similar to Eq. (1). The effects observed in this analysis are quite muted. The coefficients on *Big 3 Direct Ownership, Big 3 Common Ownership*, and *Other Common Ownership* are insignificant in all columns. *Other Direct Ownership* is weakly negatively related to *Industry Index*. Overall, the results suggest that these institutions are unlikely to alter RPE reference group choice at the firms they own, and there is no evidence that common ownership has any effect.

Table 8 investigates the possibility that investors alter peer groups to exclude commonly held competitors as in Eq. (3). Column (1) reports a small yet positive effect at the 10 % level of significance of *Big 3 Common Block* on RPE peer selection. This implies that firms are, if anything, more likely to select peers with whom they share a Big Three blockholder, which is contrary to the idea that the Big Three avoid pitting their portfolio firms against each other. Also in Column (1), the direct effects of the two competitiveness measures, *Product Similarity* and *Return Correlation* (in Panels A and B, respectively), are positive, meaning that firms benchmark against peers that are closer rivals in terms of product market space and economic exposure.

The remaining columns reveal that the effects attributed to the competitiveness measures are largely due to the Big Three's presence. Introducing the interactions of the direct and common ownership with *Product Similarity* and *Return Correlation* produces large effects on their interactions with *Big 3 Direct Ownership* and causes their direct effects to shrink to near zero. In Columns (2) and (6), the interaction terms involving *Big 3 Direct Ownership* are statistically positive, while *Product Similarity* and *Return Correlation* have negligible effects. These results suggest that the Big Three influence pairwise peer selection by encouraging the firms they own to benchmark performance-contingent pay against closely related rivals.

Furthermore, the coefficients on the interaction terms involving *Big 3 Common Block* and *Other Common Block* are statistically indifferent from zero. For example, the point estimate on *Big 3 Common Block* × *Product Similarity* in Column (2) of Panel A is 0.031 and its standard error is 0.022. The one-sided lower 95 % bound on this estimate is −0.004 (=0.031–1.645 × 0.022), which means that a one-sigma increase in *Product Similarity* produces a small 0.15 percentage point reduction in peer selection probability (= − 0.004 × 0.331) at firms with a *Big 3 Common Block*, or about 2.3 % of the sample mean (=0.0015/0.065). Alternatively, the same lower bound on *Big 3 Common Block* × *Return Correlation* is −0.003 (=0.000–1.645 × 0.002), which predicts an even smaller effect in conjunction with *Return Correlation*'s smaller standard deviation. Interpreting the effect of *Other Common Block* × *Product Similarity* in Column (4) of Panel A, the lower 95 % confidence bound is −0.017 (=0.029–1.645 × 0.028), where a one-sigma increase in *Other Common Block* corresponds to a 0.3 % (= − 0.017 × 0.192) reduction in peer selection probability, and the effect in Panel B is even smaller.

I conclude that common ownership does not materially affect pairwise peer selection. This statistically insignificant non-result is inconsistent with the hypothesis that institutional investors discourage competition among their portfolio companies.

4.4. Robustness tests

This section outlines several sets of robustness tests that corroborate the paper's main results and, for brevity, are reported in the Internet Appendix. Please see the Internet Appendix for complete discussions of these tests. First, I adopt the modified Herfindahl-Hirschman Index developed by O'Brien and Salop (2000) and use *ΔMHHI* – which captures common ownership by all institutional owners – as the measure of common ownership. Second, I use alternative industry specifications based on product descriptions by Hoberg and Phillips (2010, 2016). Third, I use an alternative definition of the common ownership based on relative portfolio weights as in Harford et al. (2011). Last, I estimate nonlinear alternatives to the paper's linear regression models. In all cases, the results support the main paper. Altogether, I find no evidence that common owners reduce RPE in executive compensation.

5. Conclusion

The Big Three asset management firms have significant economic power in U.S. capital markets, accounting for over 20 % of holdings at most firms in recent years (Fig. 1). These institutions have been at the center of a debate on antitrust regulations, as they regularly command stakes in natural product market competitors and could gain from collusion. Scholars including Elhauge (2016, 2021) and Posner et al. (2016) have recommended antitrust enforcement actions to limit common ownership such as divestitures or prohibiting funds from owning multiple stocks within concentrated industries.

My paper looks for evidence of anticompetitive behavior in executive compensation design and tests whether the Big Three reduce incentive programs that reward executives for outperforming their rivals. In contrast, I find that the Big Three encourage RPE at the firms they own to align pay with performance despite its known competitive effects. Their common ownership does not diminish this propensity, and my statistical estimates of these effects are tightly within the neighborhood of zero. My results do not support the hypothesis that the Big Three have anti-competitive effects and suggest that antitrust intervention may be imprudent.

CRediT authorship contribution statement

Thomas Schneider: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Appendix A. Variable definitions

Relative Performance Evaluation

RPE: A dummy variable that indicates if the firm uses relative performance evaluation as a component of CEO compensation [Incentive Lab]
RPE%: The dollar-weighted fraction of the CEO's pay that is linked to relative performance awards [Incentive Lab]

Ownership and Instruments

Institutional Direct Ownership: The fraction of the firm's stock owned by institutional investors [Thomson Reuters]
Institutional Common Ownership: The fraction of same-industry firms that share at least one institutional investors as a blockholder with the firm [Thomson Reuters]
Big 3 Direct Ownership: The fraction of the firm's stock owned by BlackRock, State Street, and Vanguard [Thomson Reuters]
Big 3 Common Ownership: The fraction of same-industry firms that share at least one of the Big Three as a blockholder with the firm [Thomson Reuters]
Other Direct Ownership: The fraction of the firm's stock owned by institutional investors, excluding BlackRock, State Street, and Vanguard [Thomson Reuters]
Other Common Ownership: The fraction of same-industry firms that share at least one non-Big Three institutional investor as a blockholder with the firm [Thomson Reuters]
Firm in S&P 500: A dummy variable that indicates that the firm is in the S&P 500 [CRSP]
Industry in S&P 500: The fraction of same-industry firms that are in the S&P 500 [CRSP]
Industry ΔMHHI: The incremental increase in industry concentration attributable to common ownership as in O'Brien and Salop (2000) [CRSP]
Industry HHI: Herfindahl-Hirschman Index of industry concentration [CRSP]

Firm Characteristics

Industry Exposure: The correlation of the firm's stock returns with the industry's return over the prior year [CRSP]
Stock Return: The firm's total stockholder return including dividends [CRSP]
Return-on-Assets: Net income divided by total assets [Compustat]
Book-to-Market: Book value of equity divided by market value of equity [Compustat]
Leverage: Total debt divided by assets [Compustat]
log MCAP: Natural log of market capitalization [Compustat]
Multi-segment: A dummy if the firm is a multi-segment conglomerate [Compustat]

Executive Characteristics

CEO/Chair Duality: A dummy variable if the CEO is chairperson of the board [BoardEx, Incentive Lab]
General Ability Index: The extent to which a CEO has general managerial skills, in contrast to specialist skills, as in Custódio et al. (2013) [BoardEx]
Executive Age: The CEO's age [BoardEx, Incentive Lab]
Executive Tenure: The number of years the CEO has been in that position [BoardEx, Incentive Lab]

Say-on-Pay

Votes: The fraction of votes in support of the firm's executive compensation program, defined as votes in favor divided by the prescribed base, which is either votes in favor plus votes against, votes in favor plus votes against plus abstentions, or the number of outstanding shares [ISS Voting Analytics]
log Total Compensation: The natural log of the total fair value of CEO compensation [Incentive Lab]
Pct Stock-based Pay: The dollar-weighted fraction of the CEO's pay that is delivered in stock-based or option-based pay [Incentive Lab]

Benchmark Group

Relative to Peer Group: A dummy variable that indicates if the firm benchmarks RPE relative to a peer group [Incentive Lab]
Relative to Industry Index: A dummy variable that indicates if the firm benchmarks RPE relative to an industry index such as the S&P 500 Industrials Index [Incentive Lab]
Relative to Market Index: A dummy variable that indicates if the firm benchmarks RPE relative to an industry index such as the S&P 500 Index [Incentive Lab]

Peer Selection

Product Similarity: The Hoberg and Phillips (2010, 2016) pairwise text-based product market similarity score from 10 K product descriptions [https://hobergphillips.tuck.dartmouth.edu/]
Return Correlation: Pairwise stock return correlation between the firm and the potential peer [CRSP]
Similar Variable: A dummy variable that indicates that the firm and the peer are within one sample standard deviation of each other for a variety of metrics (MCAP, Leverage, *etc.*) [Compustat]

Appendix B. Excerpt of voting guidelines

This appendix presents an excerpt from Vanguard's 2024 U.S. Proxy Voting Policy as it pertains to advisory votes on executive compensation. The excerpt shows that Vanguard's policy exhibits a preference for the use of RPE in executive incentives. Source: https://corporate.vanguard.com/content/dam/corp/advocate/investment-stewardship/pdf/policies-and-reports/us_proxy_voting_policy_2024.pdf

Advisory votes on executive compensation (Say on Pay)

Because norms and expectations vary by industry type, company size, company age, and geographic location, the following guidelines illustrate elements of effective executive compensation plans and are not a "one-size-fits-all" tool.

A fund's considerations when evaluating executive pay fall into three broad categories:

Alignment of pay and performance. The funds look for evidence of clear alignment between pay outcomes and company performance. This is mainly assessed through alignment of incentive targets with corporate strategy and analysis of three-year total shareholder return and realized pay over the same period *vs.* a relevant set of peer companies. If there are concerns that pay and performance are not aligned, a fund may vote against a pay-related proposal.

Compensation plan structure. Plan structures should be aligned with the company's stated long-term strategy and should support pay-for-performance alignment. Where a plan includes structural issues which the funds determine have led to, or could in the future lead to, pay-for-performance misalignment, a fund may vote against a pay-related proposal. For compensation structures which are not typical of a market, the Vanguard-advised funds look for specific disclosure demonstrating how the structure supports long-term value creation for shareholders.

Governance of compensation plans. The funds look for boards to have a clear strategy and philosophy on executive pay, utilize robust processes to evaluate and evolve executive pay plans, and implement executive pay plans responsive to shareholder feedback over time. The funds also look for boards to explain these matters to shareholders *via* company disclosures. Where pay-related proposals consistently receive low support, the funds look for boards to demonstrate responsiveness to shareholder concerns.

A fund will vote case by case on executive compensation proposals (including Say on Pay, compensation reports, and compensation policies) and in general will support those that enhance long-term shareholder value. It may also vote for compensation proposals that reflect improvements in compensation practices in the interests of long-term shareholder value creation, even if the proposals are not perfectly aligned with all these guidelines.

While a fund will not be prescriptive as to exact structure of a compensation plan, it will seek structures and processes that can reasonably be expected to align pay and performance over time. Such structures may include a meaningful portion of equity vesting on performance criteria, strategically aligned performance metrics set to rigorous goals, and clear disclosure of the program and outcomes enabling shareholders to understand the connection to long-term shareholder value, among other factors. A fund does not look for nonfinancial metrics (such as environmental, social, and governance [ESG] metrics) to be a standard component of all compensation plans. When compensation committees choose to include nonfinancial metrics, the funds look for the same qualities the funds do with more traditional metrics, such as rigor, disclosure, and alignment with key strategic goals and/or material risks.

The following situations are among those that raise a higher level of concern related to a compensation plan:

- Pay outcomes are significantly higher than those of peers but total shareholder return is well below that of peers.
- The long-term plan makes up less than 50 % of total pay.
- The long-term plan has a performance period of less than three years.
- Plan targets are reset or retested or are not rigorous.
- The target for total pay is set above the peer-group median.

The following situations are among those that raise warning signs, or a moderate level of concern:

- The peer group used to benchmark pay is not comparably aligned with the company in size or strategy.
- The plan uses absolute metrics only.
- The plan allows for positive discretion only.
- The company uses one-time (*e.g.*, retention) awards.
- The disclosure related to plan structure or payout is limited.

Where these warning signs exist, elements of strong compensation governance, such as board responsiveness and disclosure that includes data, rationale, and alternatives considered, can sometimes serve to mitigate these concerns.

Appendix C. Variance inflation factors

	VIF	VIF	VIF	VIF	VIF	VIF
	(1)	(2)	(3)	(4)	(3)	(4)
Big 3 Direct Ownership	2.00	5.80			2.23	6.62
Big 3 Common Ownership	1.77	3.66			1.83	3.70
Other Direct Ownership			1.15	3.64	1.40	4.42
Other Common Ownership			1.13	1.84	1.15	1.85
Industry HHI	1.12	12.94	1.12	12.92	1.12	12.94
Industry Exposure	1.32	2.69	1.30	2.69	1.33	2.69
Stock Return	1.26	1.63	1.26	1.63	1.27	1.63
Return-on-Assets	1.24	4.09	1.24	4.09	1.24	4.10
Book-to-Market	1.39	4.46	1.40	4.46	1.40	4.46
Leverage	1.19	6.37	1.18	6.37	1.19	6.38
log Market Cap.	1.36	13.70	1.39	13.73	1.40	13.75
Multi-segment	1.06	5.19	1.06	5.19	1.06	5.19
Current Chair	1.23	3.79	1.23	3.79	1.23	3.80
General Ability Index	1.09	4.07	1.09	4.08	1.10	4.08
Executive Age	1.31	4.78	1.32	4.77	1.32	4.78
Executive Tenure	1.36	4.37	1.35	4.36	1.36	4.37
Firm FE	No	Yes	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	11,516	11,426	11,516	11,426	11,516	11,426

This table reports variance inflation factors (VIF) for the regression models involving ownership and common ownership. VIF values below the commonly accepted threshold of 10 indicate that multicollinearity is within an acceptable range. *Big 3 Direct Ownership* is the fraction of shares owned by BlackRock, State Street, and Vanguard. *Big 3 Common Ownership* is the fraction of same-industry firms that share a common Big Three blockholder with the firm. *Other Direct Ownership* is the fraction of the firm's stock owned by institutional investors excluding the Big Three. *Other Common Ownership* is the fraction of same-industry firms that share at least one non-Big Three institutional investor as a blockholder with the firm. Variables are defined in Appendix 1. Estimation is OLS with fixed effects.

Appendix D. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jcorpfin.2025.102806.

Data availability

The authors do not have permission to share data.

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